Revised empirical analysis

September 13, 2026. Canonical report accompanying the revised main paper and technical supplement.

Data and dependency accounting

The original study contains 1,084,640 saved datasets in 318 stochastic configurations plus 84 analytic-only configurations. All 303 interval-producing cases were re-evaluated from their original raw draws with the four-ULP policy and R1 scalar estimator. The remaining 15 event-only cases have no affected intervals. This is deterministic method reevaluation, not 1,084,640 new draws. All original files are preserved. The revision adds **87,200 new independent datasets**, 55 configurations, and 600 binary optimizer solves on 200 observations. Three confidence levels reuse observations, and their losses reuse an explicitly identified 95% scalar estimate where relevant; they are not independent datasets.

Case-local seeds use master seed 2026091309, SHA-256 case identifiers, trial and stream. Original saved experiments retain seed 2026091203. Configurations were registered before execution, with no outcome tuning. One thread was used within each independent simulation process; paired procedures share data and pilot randomness. Fixed-M and norm-comparable-M branches are labeled separately. Raw new outputs, the exactly derived pure-fallback ablation, and independently recomputed summaries are kept separately.

Coverage and scalar corrections

Two corrected 90% rows, both S03_007 (known-noise XCH and support oracle), change from 134/150 to 133/150. They correspond to one genuine Gaussian miss, not two independent failures. Strict coverage and four-ULP coverage are both retained; 24 strict/tolerant discrepancies at exact structural boundaries were previously verified as harmless algebraic rounding. A target-proportional tolerance is no longer used.

The S03_002 midpoint MSE changes from 12,084,661.378502 to 0.147776 for the R1 estimator, on exactly the same 500 observations. Its MSE standard error is 0.024743, and its proved noise-scale bound is 9.600647. One original sentinel event is retained, along with all old losses. All 303 cases have unchanged interval endpoints. The R1 bound is marked inapplicable when the supplied cap or structural range is violated.

New experiment design

Suite	Review issue	Design	New datasets
V01	F02, F05, F06	Fixed M signal sweep; paired norm-comparable M; isolated bias, fallback, intersection and quantile diagnostics; XCH/oracle/direct	10,000
V02	F07	One fixed arbitrary loading and direction; M=3, tau=.5; tolerance .05 at 100/600/3000 nodes	200
V03	F08	18 valid whole-row corruption/noise cases, one deliberate cap violation; Gaussian, correlated and standardized t3	19,000
V04	F08	Independent residual chi-square bound; m=1/16/64, n=2/4/16, sigma=1/3; alpha0=.04, delta=.01	54,000

Suite	Review issue	Design	New datasets
V05	E01, F04	Four p=32/64 strong saturated cases, fixed and rounded growing caps; 1,000 draws each	4,000

Findings and interpretation

Budget and ablation. At amplitude two the full fixed-M interval covers 0.960; XCH covers 0.001, outside certified regularity conditions. Bias-only removal has no effect over the first eight weak/intermediate signal levels because the original fallback is selected. It cannot explain the combined diagnostic failure. Pure fallback removal, which preserves the intersections, is conservative and longer in those cases. Changing only the validation quantile reduces coverage at several levels. These comparisons isolate different mechanisms; they do not establish that every correction is necessary at every mean.

Pointwise behavior. The four new larger cases have coverage 0.965–0.978 and normalized lengths 7.20–8.01. Amplitudes deliberately satisfy the conservative P1 sufficient bound. Rounded caps and finite dimensions prevent fitting an exact asymptotic exponent from these four points. Earlier slopes and their caveats remain in the corrected archive; no favorable pooling across different regimes is performed.

Robustness. Median coverage is 1.000 for every tested within-cap row-contamination case, but interval lengths are conservative. The 120-row/zero-supplied-cap violation has coverage 0.000. The uncorrected mean also fails with correlated coordinates or row corruption. These tests address a coordinate-envelope proposition, not general tensor contamination minimaxity.

Unknown variance. The independent upper bound is conservative across the tested grid. For m=1, n=2, sigma=1, coverage is 0.9950 versus 0.7033 for the naive plug-in. At larger m, structural intervals often dominate; coverage 1 there does not demonstrate accurate variance estimation or efficient interval length. Calibration-event probabilities and Wilson intervals are separately reported.

Computation. 7/600 binary solves meet the common tolerance; 593 remain unresolved. All seven successes are at amplitude 2.5 and budget 3,000. Mean endpoint gaps at that budget range from 0.1275 to 0.3526. All observations and finite witness gaps remain in coverage, width and gap summaries. This numerical search does not settle general weak-signal computation.

Comparator and uncertainty contracts

XCH is the published tensor-PCA algorithm locally reimplemented from Xu–Chen–Han, Section 4.2, Algorithm 3; the related inference statements are in Section 4.3. HOSVD and two simultaneous updates use the same full observation as the other methods. Known-noise and plug-in variants are both retained. Zero and supplied-rank mismatch cases are outside the specified regular class. Other case-level initialization, loading-alignment and remainder constants are not certified, even if the signal is very large. The support oracle receives true support and is labeled as an information benchmark.

Coverage error bars are Wilson 95% intervals; coverage MCSE and paired coverage-difference SE are also exported. Mean-length and finite-moment loss summaries retain sample standard errors. No normal-theory MSE uncertainty is claimed for the t3 experiment: sample diagnostics are retained separately because fourth-moment arguments for the naive mean fail. Empty intervals, failures and catastrophic losses are never removed. No claim of uniform validity is inferred solely from simulated coverage.

Figures

Analytic minimax rate scales for the normalized flat loading, with unspecified multiplicative constants set to one. Left: all-budget dependence at p=4096. Right: full-data and diagonal-information scales at M=tau sqrt(p); the direct line is a noise-scale benchmark. These curves show theorem orders, not attained experimental performance or leading-constant comparisons.

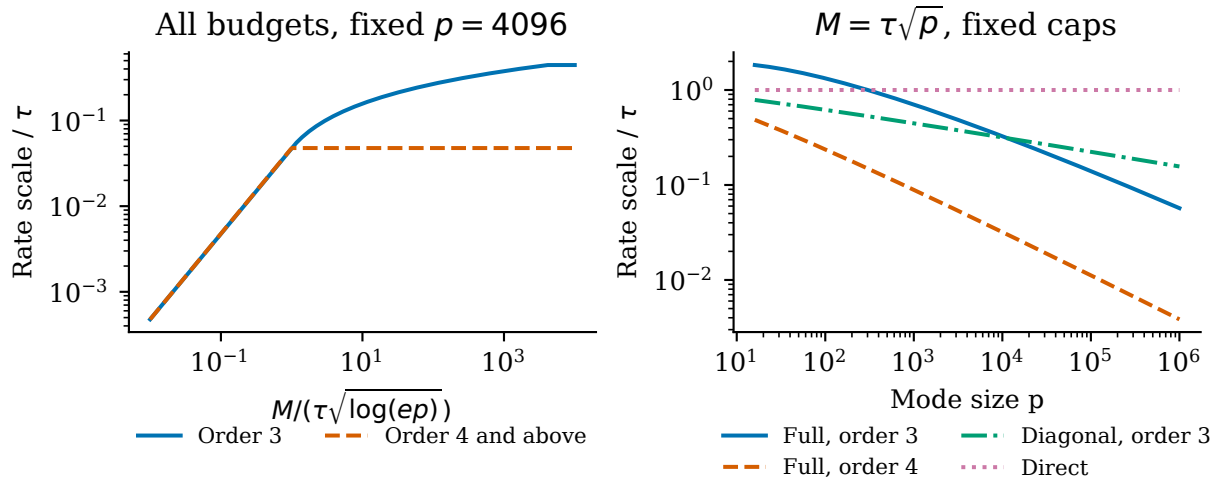


Figure 1: Analytic minimax rate scales for the normalized flat loading, with unspecified multiplicative constants set to one. Left: all-budget dependence at p=4096. Right: full-data and diagonal-information scales at M=tau sqrt(p); the direct line is a noise-scale benchmark. These curves show theorem orders, not attained experimental performance or leading-constant comparisons.

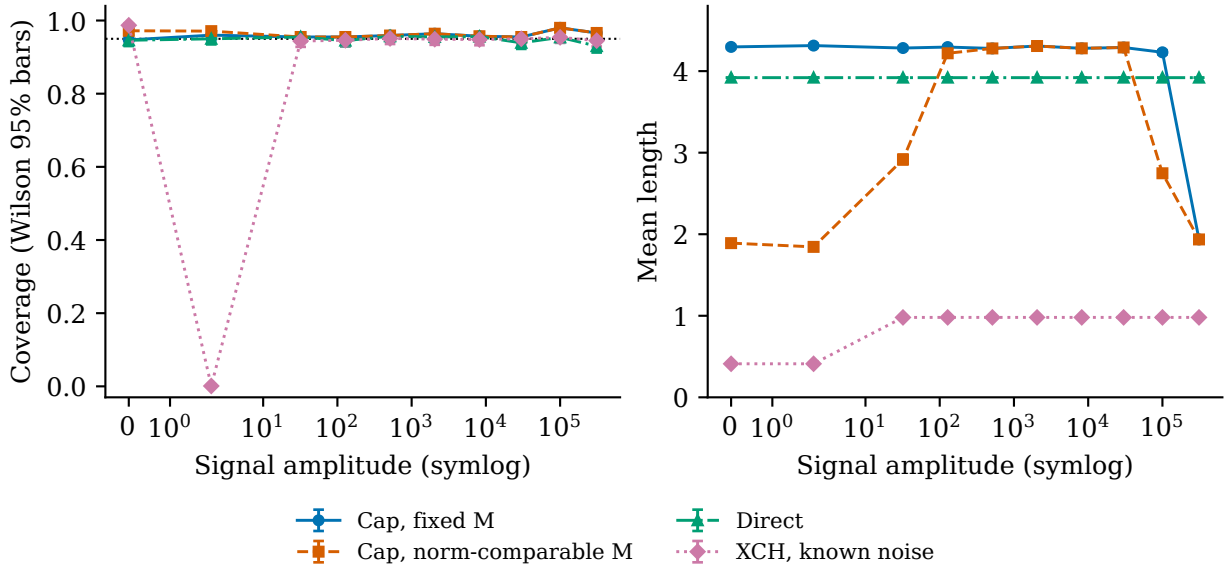


Figure 2: Paired signal sweep at p=16, rank cap one and tau=1, with 1,000 new datasets per amplitude. Fixed-M inference uses the same M=342023.69 throughout; the norm-comparable branch changes M while retaining the same observations and pilot. Bars show Wilson 95% coverage intervals or 1.96 standard errors for mean length. XCH is a regular-stratum comparator whose weak-signal conditions are not certified; it is not cap-uniform.

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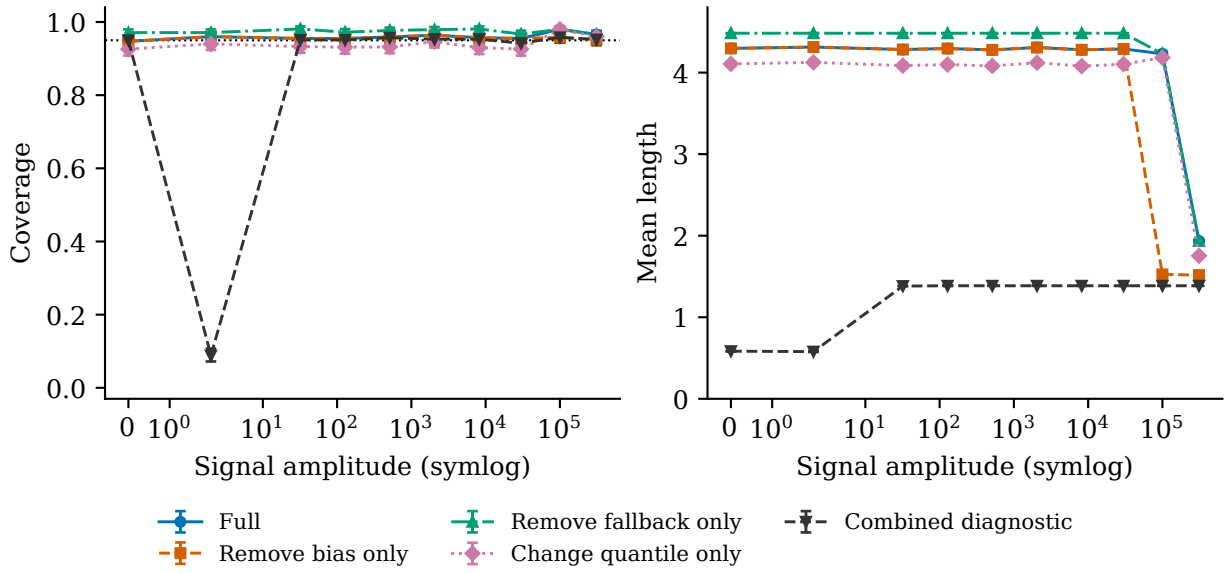


Figure 3: Isolated ablations use the same data, pilot projectors, supplied M and other calibration as the full procedure (1,000 datasets per amplitude). Removing bias freezes the original candidate choice; removing fallback forces the tangent candidate but preserves both intersections. Changing the validation quantile preserves the candidate choice, bias and intersections. The historical combined diagnostic removes multiple protections, so its weak-signal failure cannot identify the effect of bias alone; all paired differences and uncertainties are retained in the tables.

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Four new strong saturated cases use 1,000 datasets each, increasing precision at p=32 and p=64. Left: raw, unconditional losses retain all empty-intersection events; the R1 companion changes only the scalar estimate. Full MSE standard errors, sentinel counts and paired losses are in the table; a logarithmic axis displays their extreme scale difference without treating the rare-loss mean as precisely estimated. Right: finite-grid lengths normalized by the P1 rate, with 1.96 standard errors; constants and rounded growing caps preclude reading this as an exact slope claim.

Independent variance calibration uses Gaussian replicate residuals, chi-square error delta=.01 and inference error alpha0=.04; the displayed m=1, sigma=1 cases have 3,000 datasets each. Whole-replicate corruption uses the finite-variance median envelope with 1,000 datasets per case, m=16 and n=512. The separately marked 120-row case supplies a zero corruption cap and deliberately violates the theorem. All coverage bars are Wilson 95% intervals; the full m/n/noise grids and conservative lengths appear in the accompanying tables.

All budgets use the same normalized arbitrary loading, signal direction, M=3, tau=.5 and endpoint tolerance .05 in a 2 by 2 by 2 rank-one model. Each amplitude uses 50 new observations shared across the three budgets, giving 600 solves. Every unresolved global gap is retained; mean-gap bars are 1.96 standard errors, while timing is descriptive. Seven of 600 solves meet the requested tolerance, all at the strongest signal and largest budget; guarded float64 outer bounds are not formal interval arithmetic or a general high-dimensional minimax implementation.

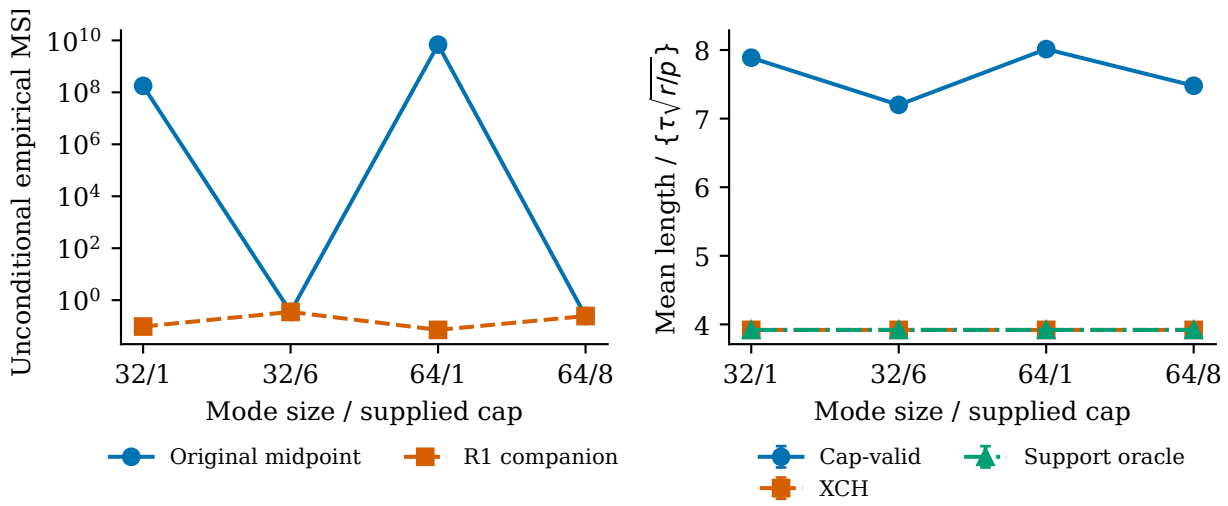


Figure 4: Four new strong saturated cases use 1,000 datasets each, increasing precision at p=32 and p=64. Left: raw, unconditional losses retain all empty-intersection events; the R1 companion changes only the scalar estimate. Full MSE standard errors, sentinel counts and paired losses are in the table; a logarithmic axis displays their extreme scale difference without treating the rare-loss mean as precisely estimated. Right: finite-grid lengths normalized by the P1 rate, with 1.96 standard errors; constants and rounded growing caps preclude reading this as an exact slope claim.

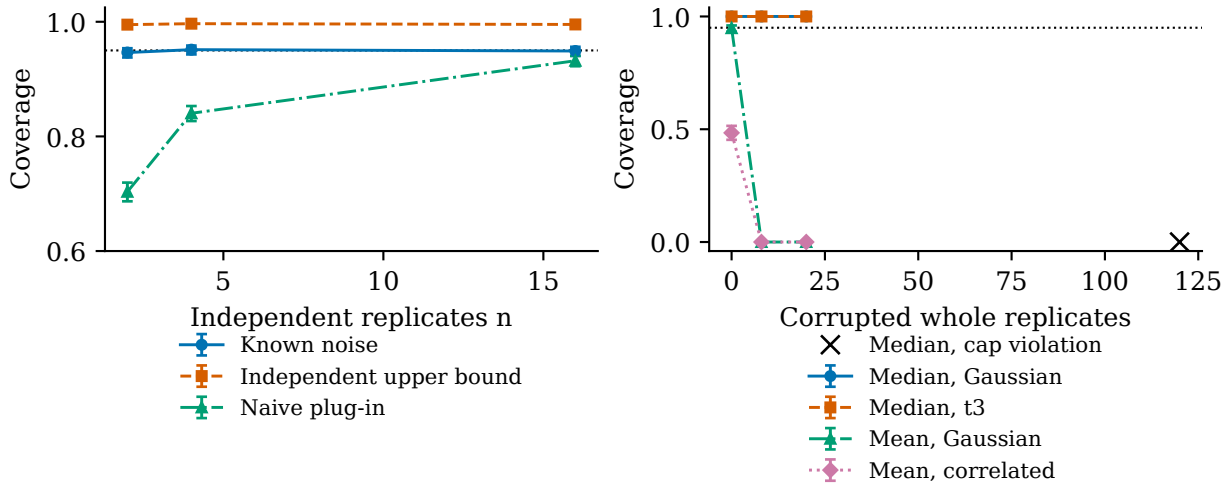


Figure 5: Independent variance calibration uses Gaussian replicate residuals, chi-square error delta=.01 and inference error alpha0=.04; the displayed m=1, sigma=1 cases have 3,000 datasets each. Whole-replicate corruption uses the finite-variance median envelope with 1,000 datasets per case, m=16 and n=512. The separately marked 120-row case supplies a zero corruption cap and deliberately violates the theorem. All coverage bars are Wilson 95% intervals; the full m/n/noise grids and conservative lengths appear in the accompanying tables.

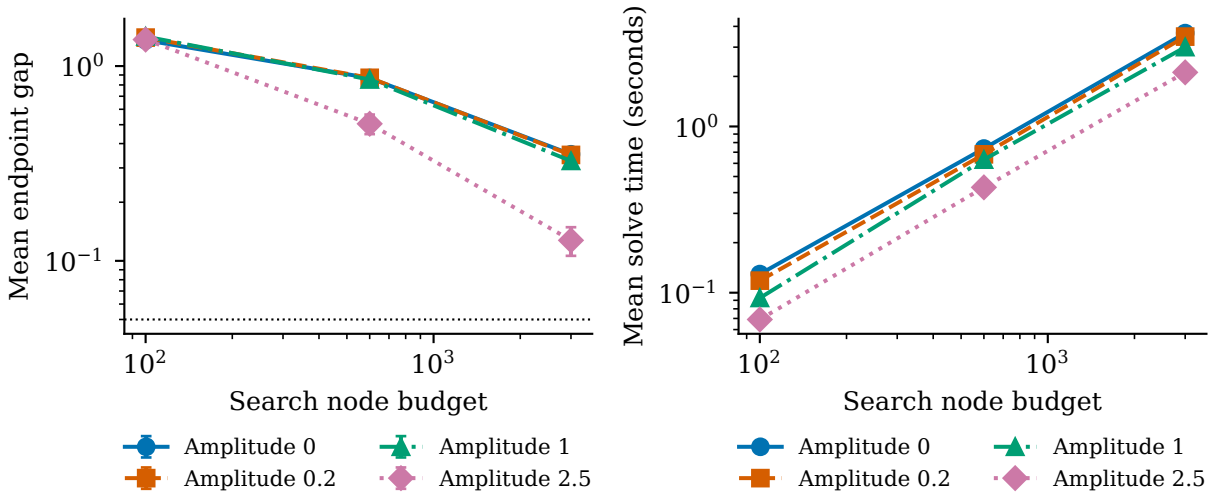


Figure 6: All budgets use the same normalized arbitrary loading, signal direction, M=3, tau=.5 and endpoint tolerance .05 in a 2 by 2 by 2 rank-one model. Each amplitude uses 50 new observations shared across the three budgets, giving 600 solves. Every unresolved global gap is retained; mean-gap bars are 1.96 standard errors, while timing is descriptive. Seven of 600 solves meet the requested tolerance, all at the strongest signal and largest budget; guarded float64 outer bounds are not formal interval arithmetic or a general high-dimensional minimax implementation.

Complete 95% coverage and length tables

All methods and confidence levels, including paired differences and estimator identities, are available in tables/revision_summary.csv. The following tables include every configuration at 95% with the principal methods; all controlled variants remain in the machine-readable table.

V01

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V01_000 / direct	0.946 [0.930, 0.958]	3.920 (0.0000)
V01_000 / cap_fixed	0.947 [0.931, 0.959]	4.297 (0.0117)
V01_000 / cap_matched	0.972 [0.960, 0.981]	1.891 (0.0087)
V01_000 / split_unprotected	0.950 [0.935, 0.962]	0.584 (0.0022)
V01_000 / xch_known	0.987 [0.978, 0.992]	0.411 (0.0015)
V01_001 / direct	0.950 [0.935, 0.962]	3.920 (0.0000)
V01_001 / cap_fixed	0.960 [0.946, 0.970]	4.313 (0.0109)
V01_001 / cap_matched	0.971 [0.959, 0.980]	1.845 (0.0110)
V01_001 / split_unprotected	0.088 [0.072, 0.107]	0.578 (0.0022)
V01_001 / xch_known	0.001 [0.000, 0.006]	0.411 (0.0014)
V01_002 / direct	0.955 [0.940, 0.966]	3.920 (0.0000)
V01_002 / cap_fixed	0.955 [0.940, 0.966]	4.283 (0.0121)
V01_002 / cap_matched	0.955 [0.940, 0.966]	2.916 (0.0306)
V01_002 / split_unprotected	0.951 [0.936, 0.963]	1.382 (0.0001)
V01_002 / xch_known	0.944 [0.928, 0.957]	0.980 (0.0000)
V01_003 / direct	0.945 [0.929, 0.958]	3.920 (0.0000)
V01_003 / cap_fixed	0.955 [0.940, 0.966]	4.295 (0.0115)
V01_003 / cap_matched	0.955 [0.940, 0.966]	4.219 (0.0133)
V01_003 / split_unprotected	0.951 [0.936, 0.963]	1.386 (0.0000)
V01_003 / xch_known	0.946 [0.930, 0.958]	0.980 (0.0000)
V01_004 / direct	0.957 [0.943, 0.968]	3.920 (0.0000)
V01_004 / cap_fixed	0.959 [0.945, 0.970]	4.278 (0.0120)

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V01_004 / cap_matched	0.959 [0.945, 0.970]	4.278 (0.0120)
V01_004 / split_unprotected	0.957 [0.943, 0.968]	1.386 (0.0000)
V01_004 / xch_known	0.951 [0.936, 0.963]	0.980 (0.0000)
V01_005 / direct	0.955 [0.940, 0.966]	3.920 (0.0000)
V01_005 / cap_fixed	0.964 [0.951, 0.974]	4.308 (0.0108)
V01_005 / cap_matched	0.964 [0.951, 0.974]	4.308 (0.0108)
V01_005 / split_unprotected	0.953 [0.938, 0.964]	1.386 (0.0000)
V01_005 / xch_known	0.949 [0.934, 0.961]	0.980 (0.0000)
V01_006 / direct	0.956 [0.941, 0.967]	3.920 (0.0000)
V01_006 / cap_fixed	0.957 [0.943, 0.968]	4.280 (0.0122)
V01_006 / cap_matched	0.957 [0.943, 0.968]	4.280 (0.0122)
V01_006 / split_unprotected	0.953 [0.938, 0.964]	1.386 (0.0000)
V01_006 / xch_known	0.948 [0.932, 0.960]	0.980 (0.0000)
V01_007 / direct	0.939 [0.922, 0.952]	3.920 (0.0000)
V01_007 / cap_fixed	0.955 [0.940, 0.966]	4.290 (0.0123)
V01_007 / cap_matched	0.955 [0.940, 0.966]	4.290 (0.0123)
V01_007 / split_unprotected	0.942 [0.926, 0.955]	1.386 (0.0000)
V01_007 / xch_known	0.950 [0.935, 0.962]	0.980 (0.0000)
V01_008 / direct	0.954 [0.939, 0.965]	3.920 (0.0000)
V01_008 / cap_fixed	0.980 [0.969, 0.987]	4.231 (0.0130)
V01_008 / cap_matched	0.980 [0.969, 0.987]	2.747 (0.0137)
V01_008 / split_unprotected	0.958 [0.944, 0.969]	1.386 (0.0000)
V01_008 / xch_known	0.954 [0.939, 0.965]	0.980 (0.0000)
V01_009 / direct	0.930 [0.912, 0.944]	3.920 (0.0000)
V01_009 / cap_fixed	0.966 [0.953, 0.976]	1.936 (0.0092)
V01_009 / cap_matched	0.966 [0.953, 0.976]	1.936 (0.0092)
V01_009 / split_unprotected	0.952 [0.937, 0.964]	1.386 (0.0000)
V01_009 / xch_known	0.945 [0.929, 0.958]	0.980 (0.0000)

V02

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V02_000 / direct	1.000 [0.929, 1.000]	1.960 (0.0000)
V02_000 / global_100	1.000 [0.929, 1.000]	6.000 (0.0000)
V02_000 / global_600	1.000 [0.929, 1.000]	5.545 (0.0198)
V02_000 / global_3000	1.000 [0.929, 1.000]	4.631 (0.0072)
V02_001 / direct	0.920 [0.812, 0.968]	1.960 (0.0000)
V02_001 / global_100	1.000 [0.929, 1.000]	6.000 (0.0000)
V02_001 / global_600	1.000 [0.929, 1.000]	5.501 (0.0223)
V02_001 / global_3000	1.000 [0.929, 1.000]	4.606 (0.0098)
V02_002 / direct	0.960 [0.865, 0.989]	1.960 (0.0000)
V02_002 / global_100	1.000 [0.929, 1.000]	5.999 (0.0005)
V02_002 / global_600	1.000 [0.929, 1.000]	5.460 (0.0296)
V02_002 / global_3000	1.000 [0.929, 1.000]	4.551 (0.0173)
V02_003 / direct	0.940 [0.838, 0.979]	1.960 (0.0000)
V02_003 / global_100	1.000 [0.929, 1.000]	5.687 (0.0559)
V02_003 / global_600	1.000 [0.929, 1.000]	4.671 (0.0739)
V02_003 / global_3000	1.000 [0.929, 1.000]	4.005 (0.0431)

V03

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V03_000 / naive_gaussian	0.948 [0.932, 0.960]	0.173 (0.0000)
V03_000 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_001 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_001 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_002 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_002 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_003 / naive_gaussian	0.955 [0.940, 0.966]	0.122 (0.0000)
V03_003 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_004 / naive_gaussian	0.000 [0.000, 0.004]	0.122 (0.0000)
V03_004 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_005 / naive_gaussian	0.000 [0.000, 0.004]	0.122 (0.0000)
V03_005 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_006 / naive_gaussian	0.953 [0.938, 0.964]	0.173 (0.0000)
V03_006 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_007 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_007 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_008 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_008 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_009 / naive_gaussian	0.953 [0.938, 0.964]	0.122 (0.0000)
V03_009 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_010 / naive_gaussian	0.000 [0.000, 0.004]	0.122 (0.0000)
V03_010 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_011 / naive_gaussian	0.000 [0.000, 0.004]	0.122 (0.0000)
V03_011 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_012 / naive_gaussian	0.484 [0.453, 0.515]	0.173 (0.0000)
V03_012 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_013 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_013 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_014 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_014 / bounded_row_median	1.000 [0.996, 1.000]	5.210 (0.0000)
V03_015 / naive_gaussian	0.476 [0.445, 0.507]	0.122 (0.0000)
V03_015 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_016 / naive_gaussian	0.000 [0.000, 0.004]	0.122 (0.0000)
V03_016 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_017 / naive_gaussian	0.000 [0.000, 0.004]	0.122 (0.0000)
V03_017 / bounded_row_median	1.000 [0.996, 1.000]	4.472 (0.0000)
V03_018 / naive_gaussian	0.000 [0.000, 0.004]	0.173 (0.0000)
V03_018 / bounded_row_median	0.000 [0.000, 0.004]	5.210 (0.0000)

V04

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V04_000 / known_sigma	0.946 [0.938, 0.954]	2.772 (0.0000)
V04_000 / independent_upper_sigma	0.995 [0.992, 0.997]	7.873 (0.0153)
V04_000 / naive_plugin_sigma	0.703 [0.687, 0.719]	2.229 (0.0305)
V04_001 / known_sigma	1.000 [0.999, 1.000]	8.000 (0.0000)
V04_001 / independent_upper_sigma	0.996 [0.993, 0.997]	7.963 (0.0081)
V04_001 / naive_plugin_sigma	0.687 [0.671, 0.704]	5.086 (0.0505)
V04_002 / known_sigma	0.951 [0.943, 0.958]	1.960 (0.0000)
V04_002 / independent_upper_sigma	0.997 [0.994, 0.998]	7.050 (0.0290)

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V04_002 / naive_plugin_sigma	0.840 [0.827, 0.853]	1.773 (0.0140)
V04_003 / known_sigma	0.952 [0.943, 0.959]	5.880 (0.0000)
V04_003 / independent_upper_sigma	0.999 [0.998, 1.000]	7.963 (0.0062)
V04_003 / naive_plugin_sigma	0.854 [0.841, 0.866]	5.259 (0.0348)
V04_004 / known_sigma	0.949 [0.941, 0.956]	0.980 (0.0000)
V04_004 / independent_upper_sigma	0.995 [0.992, 0.997]	1.708 (0.0058)
V04_004 / naive_plugin_sigma	0.932 [0.922, 0.940]	0.963 (0.0032)
V04_005 / known_sigma	0.942 [0.933, 0.950]	2.940 (0.0000)
V04_005 / independent_upper_sigma	0.996 [0.993, 0.997]	5.149 (0.0172)
V04_005 / naive_plugin_sigma	0.923 [0.913, 0.932]	2.902 (0.0097)
V04_006 / known_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_006 / independent_upper_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_006 / naive_plugin_sigma	0.993 [0.989, 0.995]	1.987 (0.0010)
V04_007 / known_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_007 / independent_upper_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_007 / naive_plugin_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_008 / known_sigma	1.000 [0.999, 1.000]	1.930 (0.0000)
V04_008 / independent_upper_sigma	1.000 [0.999, 1.000]	1.998 (0.0002)
V04_008 / naive_plugin_sigma	0.967 [0.960, 0.973]	1.882 (0.0022)
V04_009 / known_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_009 / independent_upper_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_009 / naive_plugin_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_010 / known_sigma	0.952 [0.944, 0.959]	0.980 (0.0000)
V04_010 / independent_upper_sigma	0.976 [0.970, 0.981]	1.146 (0.0010)
V04_010 / naive_plugin_sigma	0.950 [0.942, 0.958]	0.978 (0.0008)
V04_011 / known_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_011 / independent_upper_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_011 / naive_plugin_sigma	1.000 [0.999, 1.000]	2.000 (0.0000)
V04_012 / known_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_012 / independent_upper_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_012 / naive_plugin_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_013 / known_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_013 / independent_upper_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_013 / naive_plugin_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_014 / known_sigma	1.000 [0.999, 1.000]	0.997 (0.0000)
V04_014 / independent_upper_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_014 / naive_plugin_sigma	1.000 [0.999, 1.000]	0.991 (0.0002)
V04_015 / known_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_015 / independent_upper_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_015 / naive_plugin_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_016 / known_sigma	1.000 [0.999, 1.000]	0.791 (0.0000)

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V04_016 / independent_upper_sigma	1.000 [0.999, 1.000]	0.809 (0.0001)
V04_016 / naive_plugin_sigma	1.000 [0.999, 1.000]	0.791 (0.0001)
V04_017 / known_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_017 / independent_upper_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)
V04_017 / naive_plugin_sigma	1.000 [0.999, 1.000]	1.000 (0.0000)

V05

Case / method	Coverage [Wilson 95%]	Mean length (SE)
V05_000 / direct	0.937 [0.920, 0.950]	3.920 (0.0000)
V05_000 / cap_fixed	0.965 [0.952, 0.975]	1.394 (0.0061)
V05_000 / xch_known	0.946 [0.930, 0.958]	0.693 (0.0000)
V05_000 / oracle	0.946 [0.930, 0.958]	0.693 (0.0000)
V05_001 / direct	0.953 [0.938, 0.964]	3.920 (0.0000)
V05_001 / cap_fixed	0.977 [0.966, 0.985]	3.118 (0.0159)
V05_001 / xch_known	0.949 [0.934, 0.961]	1.697 (0.0000)
V05_001 / oracle	0.949 [0.934, 0.961]	1.697 (0.0000)
V05_002 / direct	0.959 [0.945, 0.970]	3.920 (0.0000)
V05_002 / cap_fixed	0.978 [0.967, 0.985]	1.002 (0.0041)
V05_002 / xch_known	0.955 [0.940, 0.966]	0.490 (0.0000)
V05_002 / oracle	0.955 [0.940, 0.966]	0.490 (0.0000)
V05_003 / direct	0.950 [0.935, 0.962]	3.920 (0.0000)
V05_003 / cap_fixed	0.966 [0.953, 0.976]	2.644 (0.0133)
V05_003 / xch_known	0.953 [0.938, 0.964]	1.386 (0.0000)
V05_003 / oracle	0.953 [0.938, 0.964]	1.386 (0.0000)

Optimizer precision

Case	Nodes	Certified / 50	Mean gap (SE)	Seconds
V02_000	100	0	1.3572 (0.0205)	0.1294
V02_000	600	0	0.8681 (0.0086)	0.7367
V02_000	3000	0	0.3526 (0.0055)	3.6437
V02_001	100	0	1.4000 (0.0272)	0.1182
V02_001	600	0	0.8704 (0.0132)	0.6797
V02_001	3000	0	0.3498 (0.0087)	3.4713
V02_002	100	0	1.4175 (0.0300)	0.0929
V02_002	600	0	0.8529 (0.0146)	0.6313
V02_002	3000	0	0.3258 (0.0102)	3.0005
V02_003	100	0	1.3675 (0.0362)	0.0690
V02_003	600	0	0.5055 (0.0297)	0.4303
V02_003	3000	7	0.1275 (0.0109)	2.1160

Input and allocation accounting

Fresh timing of the revised public API: one warmup plus ten calls; one BLAS thread. Trace excludes preallocated input and may exclude native workspace, so it is not process peak memory. Coordinate-access bytes describe the separate diagonal method, not the full-tensor method.

p	Input bytes	Diagonal access bytes	API seconds (SE)	Peak traced call bytes
8	4096	64	0.000765 (0.000026)	44053
16	32768	128	0.001512 (0.000068)	299013
32	262144	256	0.009218 (0.000173)	2167429
64	2097152	512	0.087627 (0.001057)	16848773

Real-data scope

No suitable repeated-tensor dataset and prescribed scientific contrast was supplied or established. The independent unit, dependence, known-bound justification and preprocessing would need a separate defensible design. This revision makes no real-data or LLM application claim.

Reproducibility

Run reproduce.sh from this revision directory to rebuild analysis and documents from preserved/new raw draws. Use –rerun-new to execute the 55 registered configurations again. The original saved simulations are regenerated only by their separately preserved original pipeline; the deterministic correction then applies to all saved observations. Source hashes, environment information and validation logs distinguish generation, reevaluation and prior checks.